

7 The curve C has equation $y = \frac{2x - 3}{2x^2 - 7x - 4}$.

- (a) Find the equations of the asymptotes of C . [2]

[illegible]

- (b) Find the coordinates of any stationary points on C . [4]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- (c) Sketch C stating the coordinates of the intersections with the axes [3]

- (d) Sketch the curve with equation $y = \left| \frac{2x^2 - 5x}{2x^2 - 7x - 4} \right|$. [1]

- (e) Find in exact form the set of values of x for which $\left| \frac{2x^2 - 5x}{2x^2 - 7x - 4} \right| < \frac{1}{9}$. [5]

[illegible]

Additional page

If you use the following lined page to complete the answer(s) to any question(s), the question number(s) must be clearly shown.

[illegible]